Science Test 1 Full curriculum revision

Class 1A About 80 minutes Theory and varied practice

Review the five theory pages for about 15 minutes. Then complete P1–P8 in about 65 minutes. Use short words and phrases. The examples and diagrams here differ from the test; use your knowledge on unfamiliar examples.

Theory 1 Doctors and life processes

Symptoms are changes from normal that may suggest illness. A doctor uses observations and test results as evidence, then compares this evidence with known illnesses. Checks include heart rate and temperature; blood and urine can also be tested. More than one illness may cause the same symptom.

Ancient Egyptian doctors checked hearts and recorded treatments thousands of years ago. Modern doctors still check the heart but also use more advanced tests and scans. New evidence can change an earlier explanation; an old treatment is not reliable just because it was once believed to work.

Process	Meaning
Movement	Changing position or moving a body part
Reproduction	Making new organisms
Sensitivity	Detecting and responding to changes
Growth	Increasing in size
Respiration	Releasing energy from food in cells
Excretion	Removing waste made by the body or cells
Nutrition	Getting substances needed for life

MRS GREN helps you remember the seven processes. Plants can move parts of themselves, respond, grow, make food and reproduce. Animals usually move from place to place and eat food. Plants often keep growing for longer than humans, whose height usually stops increasing in adulthood.

Theory 2 Human organs

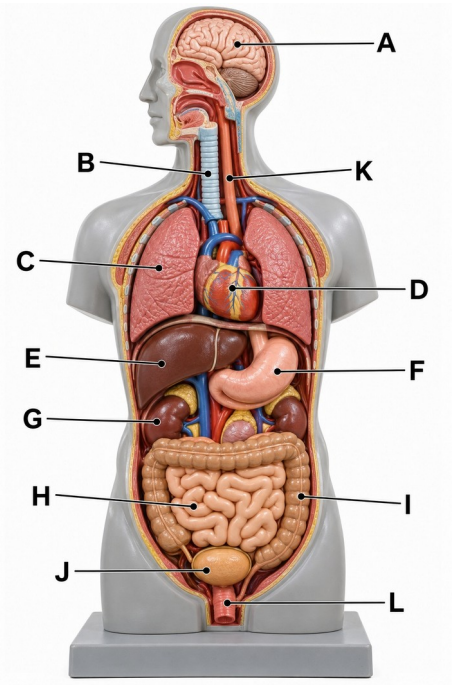


Figure key: A brain; B trachea; C lung; D heart; E liver; F stomach; G kidney; H small intestine; I large intestine; J bladder; K oesophagus; L rectum.

Organ	Job
Brain	Controls the body
Skin	Protects the body and senses changes
Trachea	Carries air towards the lungs
Oesophagus	Carries food to the stomach
Lungs	Take oxygen into blood and remove carbon dioxide
Heart	Pumps blood
Diaphragm	Muscular sheet that helps breathing
Liver	Makes and stores some substances; breaks down others
Stomach	Breaks up food
Small intestine	Digests food and absorbs nutrients
Large intestine	Absorbs water from unwanted food
Kidneys	Remove waste from blood and make urine
Rectum	Stores faeces
Bladder	Stores urine

Theory 3 Plant organs and tissues

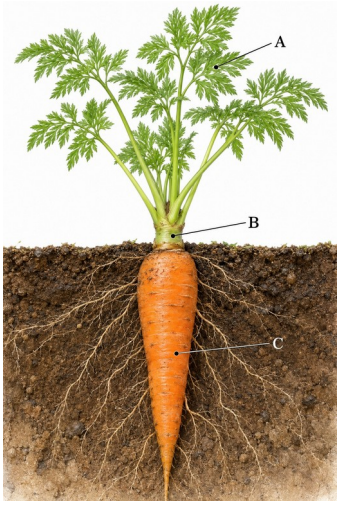


Figure key: A leaf; B stem; C carrot storage root. Leaves trap light to make food. Stems support the plant and carry substances. Roots anchor the plant and take up water and minerals. Carrots and potatoes are storage organs for food made by the plant.

Photosynthesis uses light, carbon dioxide and water to make food. Leaves need light; a plant needs roots and water as well. Animals get nutrition by eating, while green plants make food.

Cells are the basic units of life. A tissue contains cells of the same type working together. An organ contains different tissues doing different jobs. Pathologists examine organs or tissue samples to investigate disease or causes of death. A sample from a living patient is a biopsy.

Tissue	Job
Muscle	Contracts to cause movement, such as pumping blood or pushing food
Fat around the heart	Helps protect the heart
Root hair tissue	Takes up water from soil
Xylem	Carries water through roots, stems and leaves

A tissue can be useful without causing movement. For example, fat protects and xylem transports. Muscle tissue also moves the tongue and pushes food through intestines.

Theory 4 Microscopes

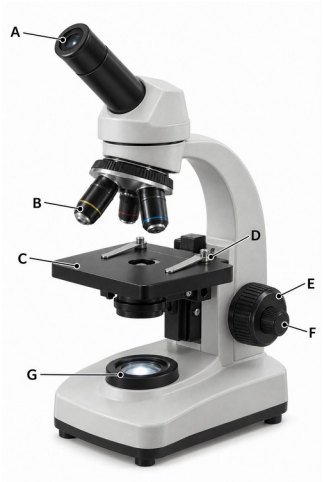


Figure key: A eyepiece; B objective; C stage; D stage clip; E coarse focus; F fine focus; G light source.

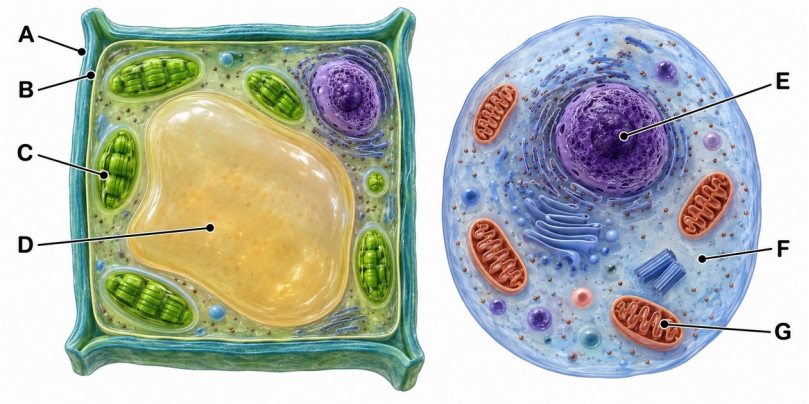
Part	Job
Eyepiece lens	Lens you look through; magnifies
Objective lens	Lens near the specimen; magnifies
Stage	Supports the slide
Stage clips	Hold the slide in place
Coarse focus	Makes large focusing changes at low power
Fine focus	Makes small focusing changes; use at higher power
Light source	Sends light through the specimen

Start at the lowest objective power: its wider view helps you find the specimen. Place and secure the slide. Adjust light and focus slowly. At higher power use fine focus; coarse movement can break the slide or damage the lens. If the view stays unclear, return to lower power. Never point a microscope mirror at the Sun.

Prepare a slide: add a drop of water, place thin tissue in it, then lower a coverslip slowly at an angle. Thin tissue lets light pass through. A coverslip holds the specimen flat/in place and helps stop drying. A stain makes structures easier to see; slow lowering avoids air bubbles.

Total magnification = eyepiece power × objective power. Example: ×8 and ×5 give ×40. A ×40 image appears 40 times larger in length; it does not contain 40 copies of the specimen.

Theory 5 Cells



The plant cell is on the LEFT in this revision figure. A wall; B membrane; C chloroplast; D vacuole; E nucleus; F cytoplasm; G mitochondrion.

Part	Job and location
Nucleus	Controls the cell; animal and plant cells
Cell membrane	Controls entry and exit; animal and plant cells
Cytoplasm	Jelly where most cell activities happen; both
Mitochondria	Release energy by respiration; both
Cell wall	Strong cellulose support; plant cells
Large vacuole	Stores cell sap; typical plant cells
Chloroplasts	Contain chlorophyll and make food by photosynthesis; green plant cells

Plant and animal cells share a nucleus, membrane, cytoplasm and mitochondria in the examples studied. Plant cells have cellulose walls and usually a large vacuole. Green leaf cells contain chloroplasts. A root hair’s long projection gives more surface in contact with soil, helping water uptake. Heart muscle cells are specialised to contract. Cell shapes suit their jobs.

Robert Hooke described cork cells in 1665. Better microscopes revealed more detail. Granville examined mummy tissue in 1825; a later examination in 2009 changed the suggested cause of death. Light microscopes magnify using lenses; electron microscopes reveal much finer detail.

Image length = real length × magnification. Real length = image length ÷ magnification. Use the same units. For example, 6 mm at ×300 represents 6 ÷ 300 = 0.02 mm. The test calculation uses lens powers; cell-size work is included here to review the supplied pages.

P1 Doctors and evidence 6 minutes

Circle one answer for each item. Then complete the table.

1. Which finding could help explain why someone feels unwell?

A) their favourite sport B) a temperature above their normal value C) their shoe size D) the colour of their notebook

2. Two doctors suggest different causes for the same symptoms. What would best help them decide?

A) a louder opinion B) using the oldest explanation C) further reliable observations or tests D) choosing at random

Check	What kind of evidence might it give?
Pulse / heart rate	
Blood or urine test	

3. An old medical book says onions can cure death. Circle the better reason for testing the claim:

A) An old belief needs evidence. B) A written claim must be correct.

4. Modern scans show details that an ancient doctor could not see. Give one reason this can improve a conclusion.

_____

P2 Life processes 9 minutes

1. Write the process that best fits each new example. Use each word once: movement, reproduction, sensitivity, growth, respiration, excretion, nutrition.

Example	Process
An adult bird produces chicks	
A young fish becomes longer	
Cells release energy to power an activity	
A person removes waste in urine	
A snail changes position	
A person detects a hot surface and pulls away	
A rabbit obtains useful substances by eating	

2. Match the letters to the THREE meanings below: A nutrition; B excretion; C reproduction.

Meaning	Letter
Making more organisms	
Getting useful substances	
Removing waste made by the body	

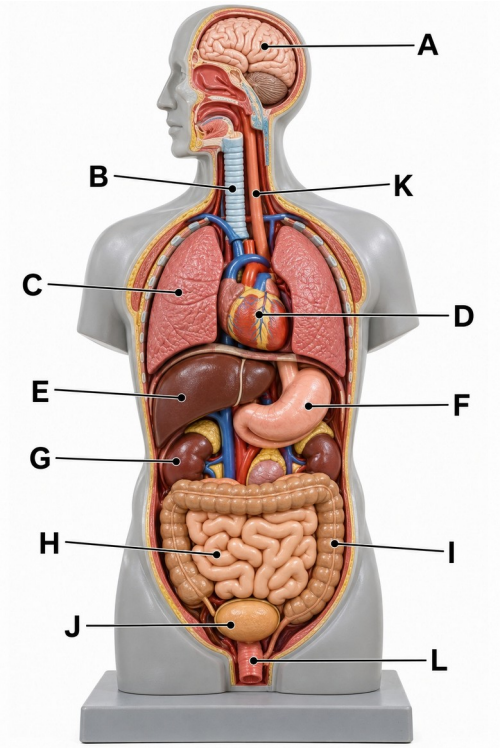
3. A stone gets larger when mud sticks to it. Does this demonstrate biological growth? Give one short reason.

Yes / No Reason: _____

4. Give ONE difference between how a fish and a tree obtain nutrition.

Fish: _____ Tree: _____

P3 Human organs 10 minutes



1. Name A–L using the new drawing. Use the organ table in Theory 2 to check afterwards.

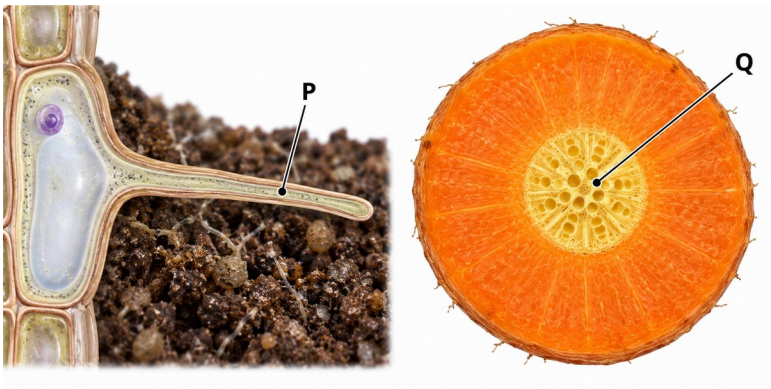
Letter	Organ	Letter	Organ
A		G	
B		H	
C		I	
D		J	
E		K	
F		L	

2. Complete each pair with one organ name. They do different jobs.

Job	Organ
Stores solid waste	
Stores liquid waste	
Takes oxygen into blood	
Removes water from unwanted food	

3. Which organ on the OUTSIDE of the body helps detect temperature and touch? _____

P4 Plants and tissues 9 minutes



1. P shows a projection on a root cell. Q shows the central tissue in a carrot root. Name them, then give their different jobs.

Letter	Name	Job
P		
Q		

2. Complete the plant-organ table with a short job.

Organ	Job
Leaf	
Stem	
Storage root	

3. Circle the three things photosynthesis needs:

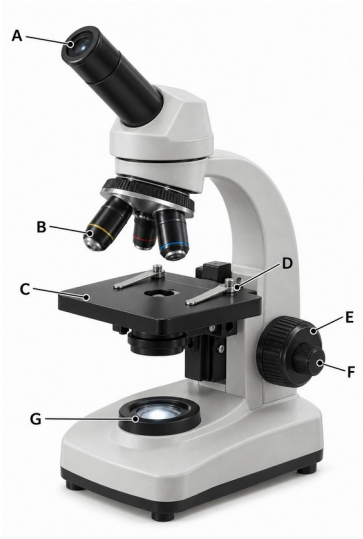
light soil as food carbon dioxide water ready-made animal food

4. Match A muscle, B fat, C xylem to each situation. Write the letter.

Situation	Letter
A tongue moves	
A layer cushions an organ	
Water moves along a leaf vein	

5. A leaf contains different tissues. Is it a cell, tissue or organ? _____

P5 Microscopes 9 minutes



1. Name A–G. This microscope is viewed from a different direction.

Letter	Part	Letter	Part
A		E	
B		F	
C		G	
D			

2. Calculate total magnification. Write the multiplication in the middle column.

Eyepiece and objective	Working	Total
×8 and ×10		
×6 and ×20		

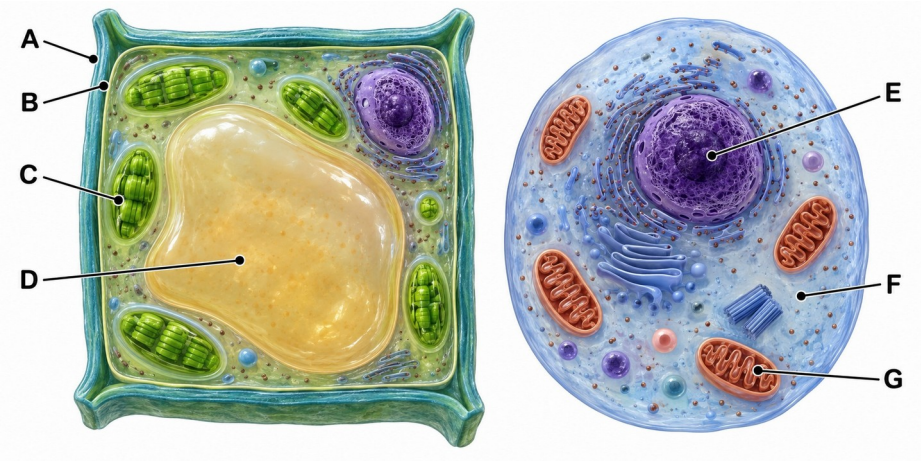
3. Put these slide-preparation steps in order by writing 1, 2 and 3:

Step	Order
Lower the coverslip slowly	
Place a drop of water on the slide	
Add a small piece of thin tissue	

4. A microscope has no slide on its stage. Is the empty stage the specimen? Give a short reason.

Yes / No Reason: _____

P6 Cells 10 minutes



1. Identify the two cells. Left: _____ Right: _____

2. Name A–G and write BOTH or PLANT for the examples studied.

Letter	Part name	Found in
A		
B		
C		
D		
E		
F		
G		

3. Name the material in a plant wall, the liquid in its large vacuole and the green pigment in a chloroplast.

Wall: _____ Vacuole: _____ Green pigment: _____

4. A student labels the thick outer box of a plant cell “membrane”. What should the label be, and where is the membrane?

Correct name: _____ Membrane position: _____

P7 Apply the science 7 minutes

Give a few words for each point. These situations practise explanations without reproducing the test questions.

1. A pond organism does not change position during a short observation. Suggest TWO other kinds of evidence you could look for to decide whether it is alive.

Evidence 1: _____ Evidence 2: _____

2. A seedling is growing towards a window. A classmate says, “Only animals can respond to surroundings.” Give one observation and what it shows.

Observation: _____ It shows: _____

3. A scientist sees a pale layer in one organ and a red contracting layer in another. Name a likely tissue for each layer and its job.

Layer	Likely tissue	Job
Pale protective layer		
Red contracting layer		

4. A pupil turns a large focusing wheel quickly while the lens is close to the slide. What could go wrong? Give a safer action.

Risk: _____ Safer action: _____

5. A cell from a pale storage organ has a wall and vacuole but no green structures. Can it still be a plant cell? Give one reason.

Yes / No Reason: _____

6. A better microscope reveals previously unseen damage in tissue. Why might a scientist revise an earlier conclusion?

_____

P8 Final review 5 minutes

1. Write the missing word in each statement.

Statement	Missing word
The basic unit of life is a ...	
Cells of the same type working together form a ...	
Different tissues working together form an ...	
Robert ... described cork cells in 1665	
A tissue sample taken from a living patient is a ...	

2. Circle the true meaning of ×250 magnification:

A) 250 new cells B) An image 250 times larger in length C) A cell 250 times smaller

3. For chapter review, complete these cell-size calculations. Use millimetres throughout.

Information	Working	Answer
Real width 0.03 mm; ×100 image		Image width: ____ mm
Image length 9 mm; magnification ×300		Real length: ____ mm

4. Tick the areas you can now explain: □ evidence □ MRS GREN □ organ jobs □ tissues □ microscope method □ cell parts. Revisit any unticked area with your teacher.